

Performance Testing

Date	3 July 2026
Team ID	xxxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Attribute	Detail
Testing Tool	Browser DevTools (Network tab), Manual test scripts
Testing Types	Functional Testing, Performance Testing, Edge Case Testing
Target Response Time	< 2 seconds from form submit to result display
Target Error Rate	< 1% on valid inputs
Live Test URL	https://credit-card-approval-prediction-ra22.onrender.com

Step 2: Test Scenarios

#	Test Scenario	Input	Expected Result	Actual Result	Status
1	Low-risk applicant	Working professional, ₹4,80,000 income, 5 years employed, no debt	Approved (high confidence)	Approved — 82% confidence	☒ Pass
2	High-risk applicant	Unemployed, ₹1,20,000 income, 0 years employed, 3 children	Rejected (high confidence)	Rejected — 79% confidence	☒ Pass
3	Mid-risk applicant	Part-time worker, ₹2,40,000 income, 1.5 years employed	Approved or Rejected (low confidence)	Approved — 56% confidence	☒ Pass
4	Form validation — missing field	Submit form without selecting income type	Browser HTML5 validation prevents submission	Validation error shown	☒ Pass
5	History page — after 3 predictions	Make 3 predictions, navigate to /history	All 3 predictions listed in reverse order	3 entries shown correctly	☒ Pass
6	History clear	Click "Clear History" on /history page	History emptied, page shows 0 entries	History cleared successfully	☒ Pass
7	Scenario pre-fill — Low Risk	Click "Low Risk" scenario button on dashboard	Prediction form fills with low-risk profile	Form filled correctly	☒ Pass

#	Test Scenario	Input	Expected Result	Actual Result	Status
8	Responsive layout — mobile	Load on 375px width device	All form fields and buttons visible without horizontal scroll	Layout correct on mobile	☒ Pass



Step 3: Performance Test Results

Metric	Target	Measured	Status
Prediction response time (warm server)	< 2 seconds	~0.3 seconds	☒ Pass
Prediction response time (cold start on Render)	< 30 seconds	~18 seconds	☒ Acceptable
Dashboard page load time	< 3 seconds	~1.2 seconds	☒ Pass
History page with 10 entries	< 2 seconds	~0.4 seconds	☒ Pass

Note: Render free tier sleeps after 15 minutes of inactivity. The first request after sleep triggers a cold start (~18 seconds). All subsequent requests are fast (<1 second).

Step 4: Model Performance Metrics (Held-Out Test Set)

Model	Accuracy	Precision (Rejected)	Recall (Rejected)	ROC-AUC
Logistic Regression	74.3%	0.71	0.68	0.81
Decision Tree	76.8%	0.73	0.74	0.77
Random Forest	81.91%	0.80	0.79	0.89
XGBoost	79.5%	0.77	0.76	0.87